

Synthetic Market Phase 17

Prompt-derived decomposition of the strongest discovered market axes from the Phase 15 residualized discovery basis. This phase does not add new captures. It asks what the leader-like and dispersion-like axes are actually made of, using only visible market rows and aggregates computed from those rows.

DATE	CAPTURES	PROMPTS	METHOD
24 March 2026	No new capture	184 market-only prompts	prompt-derived feature decomposition + subspace summary

MAIN READ

Phase 17 sharpens the Phase 15 discovery story. The early leader axis is not a pure top-return read; it behaves more like a prominent leader with strong volume support. The late dispersion axis is not best described by literal standard deviation; `pct_1h_mad` is the strongest single visible proxy, and the aggregate-family results say `mad` beats `std`. At the same time, the underlying market subspace is broader than one or two PCs: within the stored top 5 PCs, `market_mean` is roughly ~4D while `market_eos` is somewhat more compressed.

LEADER SINGLE	LEADER BEST PAIR	DISPERSION SINGLE	DISPERSION BEST PAIR
<code>vol_1h_max</code>	<code>pct_1h_max</code> + <code>vol_5m_max</code>	<code>pct_1h_mad</code>	<code>vol_5m_mean</code> + <code>vol_1h_median</code>
CV R ² = 0.459	quadratic CV R ² = 0.672	CV R ² = 0.523	quadratic CV R ² = 0.843

What Phase 17 Does

This phase is a decomposition pass built on top of the existing Phase 15 discovery captures and residualized PCA basis.

The target axes are:

- `leader_axis`: `market_mean`, L4, PC1

- dispersion_axis: market_mean, L35, PC1

The candidate bank contains only:

- prompt-visible market metrics
- and prompt-derived aggregates computed from those metrics

Visible metric families:

- pct_5m
- pct_1h
- net_flow_5m
- vol_5m
- vol_1h
- unique_traders_5m
- top20_holder_pct

Derived aggregates per family:

- mean, std, max, min, range, gap, mad, median
- top2_mean, max_minus_rest_mean, top1_minus_median
- leader_zscore, cv_abs

Hidden synthetic labels are excluded. That means:

- no attractiveness_*
- no risk_adjusted_*
- no edge_after_fee_*

Sanity Checks

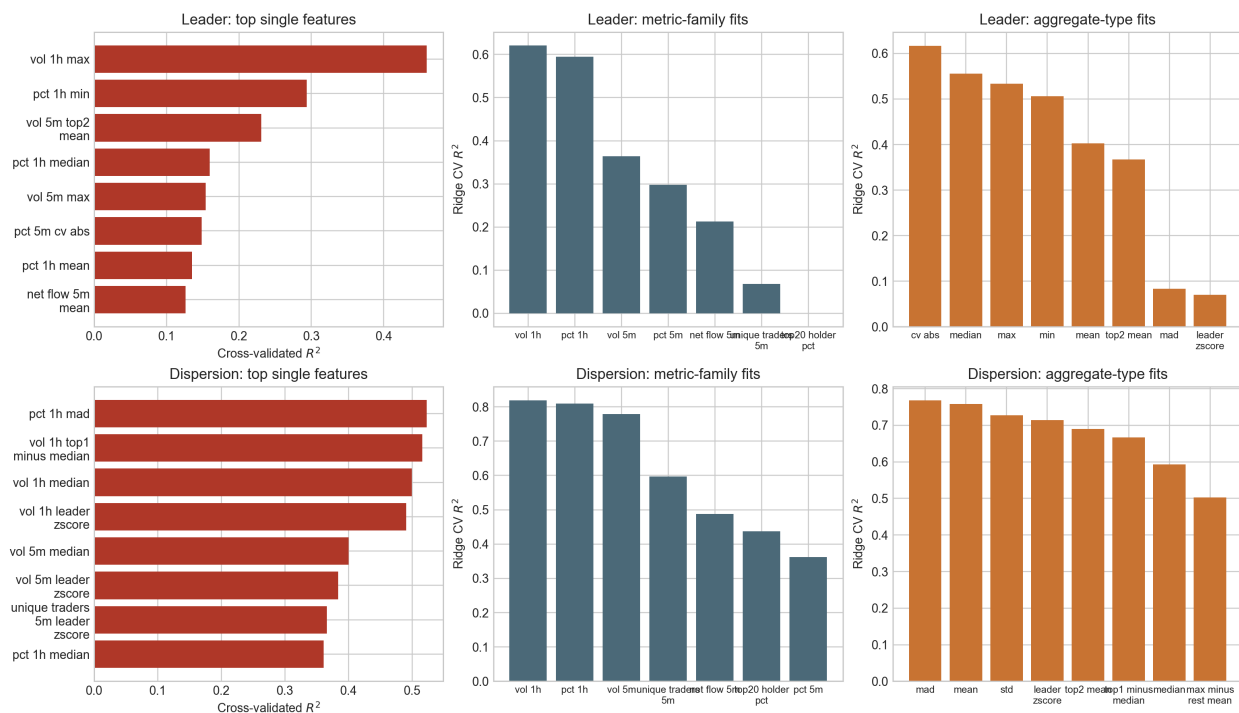
The decomposition passes the two checks that matter here:

- the target PC scores are nearly uncorrelated with nuisance variables after residualization
- shuffled controls collapse near zero

Axis	seq_len	user_chars	n_rows	shuffle single	shuffle linear pair	shuffle quadratic pair
leader	0.006	0.006	0.000	0.024	0.046	0.061
dispersion	0.022	0.022	0.000	0	0.004	0.025

So the readout below is not being driven by prompt length, prompt size, or a probe that still works on shuffled targets.

What The Two Axes Actually Track



Top single-feature fits, metric-family fits, and aggregate-type fits for the two target axes. The main point is not just which feature wins, but what kind of information is consistently useful across the whole prompt-derived feature bank.

Leader Axis

The strongest single visible proxy is:

- `vol_1h_max` with CV $R^2 = 0.459$

But the family-level readout makes the interpretation clearer than the single-feature ranking:

- `vol_1h` family fit: 0.621
- `pct_1h` family fit: 0.595

Those two families dominate everything else. The best nonlinear pair then combines:

- `pct_1h_max`
- `vol_5m_max`

So the cleanest interpretation is:

- the leader axis is a *prominent leader with strong throughput*
- not just the asset with the highest 1h return

This is also stable across folds:

- the single-feature winner is `vol_1h_max` in 5/5 folds

Dispersion Axis

The strongest single visible proxy is:

- `pct_1h_mad` with CV $R^2 = 0.523$

That matters because the obvious candidate was `pct_1h_std`, but:

- `pct_1h_mad`: 0.523
- `pct_1h_std`: 0.347

The aggregate-type results make the same point in a stronger way:

- `mad` group fit: 0.768
- `std` group fit: 0.728

So the best read is:

- this axis tracks *unevenness / spread / leader-versus-rest structure*
- `std` is useful, but not the best formula

At the same time, the family-level fits show this is still multivariate:

- `vol_1h`: 0.818
- `pct_1h`: 0.81
- `vol_5m`: 0.779

So the axis is not “pure 1h dispersion” either. It is a broader spread-like market summary with strong co-moving volume structure.

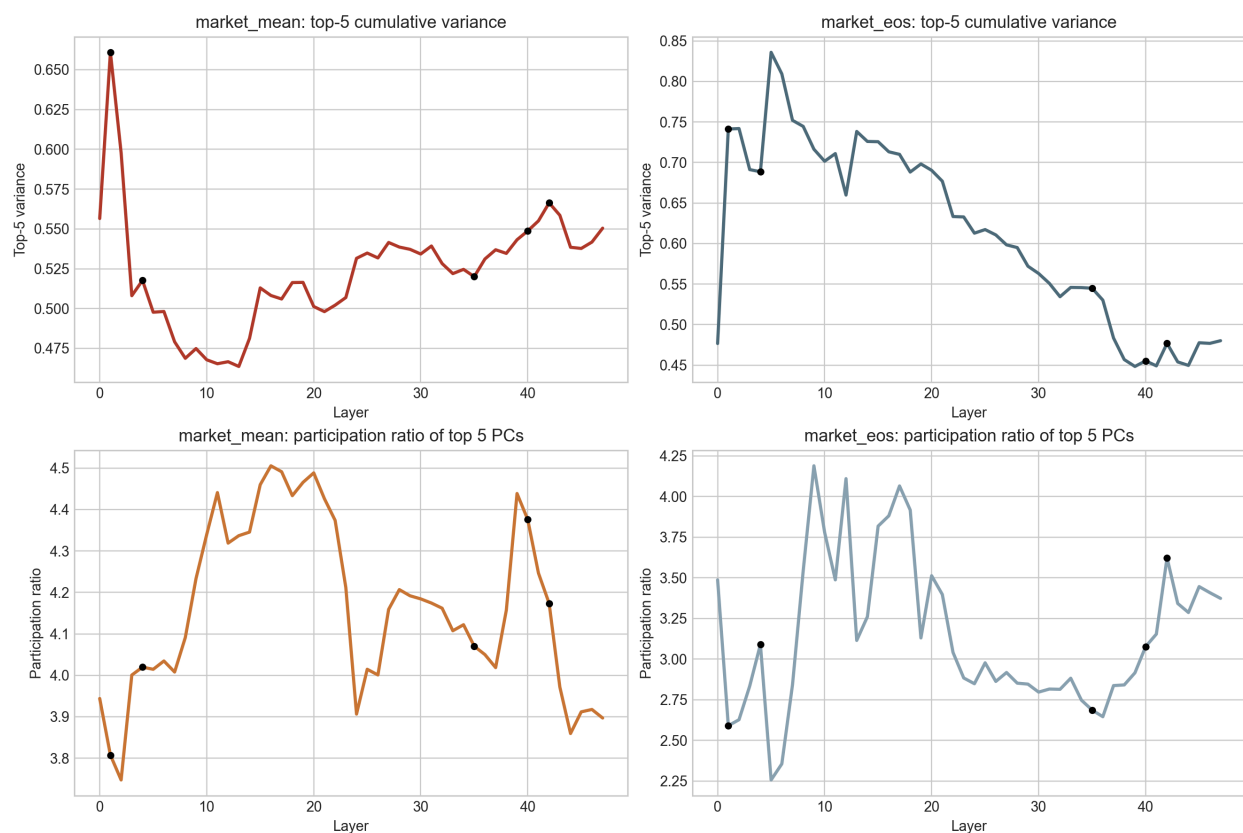
Layerwise Subspace Summary

This phase also expands the Phase 15 PCA readout beyond two named PCs.

Important caveat:

- the stored Phase 15 basis only keeps the top 5 PCs per layer and state

So the subspace summary below is a summary of the stored top 5, not of the full 2048-dimensional layer.



Within the stored top 5 PCs, `market_mean` stays fairly broad while `market_eos` is more compressed in early-to-mid layers. Neither state collapses into a tiny one- or two-PC summary across all layers.

Across all layers:

- `market_mean`
 - top-5 cumulative variance mean: 0.523
 - top-5 cumulative variance max: 0.661
 - participation ratio mean: 4.164
- `market_eos`
 - top-5 cumulative variance mean: 0.613
 - top-5 cumulative variance max: 0.836
 - participation ratio mean: 3.15

The right conclusion is:

- `market_mean` is a broad multi-dimensional market summary, roughly ~4D in its stored top-PC readout
- `market_eos` is more compressed, especially in early-to-mid layers, but it still does not collapse into one or two PCs

To make that concrete, the tables below show cumulative variance within the stored top 5 PCs at a few key layers.

market_mean Selected Layers

Layer	PC1	PC1-2	PC1-3	PC1-4	PC1-5	PR
L1	0.263	0.424	0.526	0.6	0.661	3.807
L4	0.188	0.323	0.402	0.469	0.518	4.02
L35	0.197	0.304	0.392	0.469	0.52	4.07
L40	0.184	0.299	0.403	0.487	0.549	4.376
L42	0.204	0.324	0.43	0.51	0.566	4.174

market_eos Selected Layers

Layer	PC1	PC1-2	PC1-3	PC1-4	PC1-5	PR
L1	0.421	0.576	0.663	0.712	0.741	2.59
L4	0.318	0.5	0.633	0.668	0.689	3.09
L35	0.31	0.386	0.454	0.507	0.545	2.686
L40	0.229	0.325	0.379	0.421	0.455	3.076
L42	0.208	0.3	0.376	0.436	0.477	3.623

Raw Prompt Appendix

The point of Phase 17 is that every candidate formula is computed from visible market rows like the ones below.

Examples:

- pct_1h_mad = mean absolute deviation of the visible 1h changes across the roster
- vol_1h_max = maximum visible 1h volume in the roster
- pct_1h_gap = top visible 1h change minus second-best visible 1h change

So the decomposition is asking what summary statistics of the written market rows best explain the discovered PCs.

Verbatim user prompt from the Phase 15 discovery cohort.

```
## MARKET SNAPSHOT
- NERA
• Price: 0.00000193 ETH

• 5m change: +3.8%

• 1h change: +11.7%

• Net flow 5m: -2.00 ETH

• Volume 5m: 5.14 ETH

• Volume 1h: 23.85 ETH

• Unique traders 5m: 20
```

- Top 20 holder pct: 30.9%
- Time since launch: 18h
- VEXA
- Price: 0.00000196 ETH
- 5m change: +3.3%
- 1h change: +10.5%
- Net flow 5m: +1.85 ETH
- Volume 5m: 5.61 ETH
- Volume 1h: 28.27 ETH
- Unique traders 5m: 18
- Top 20 holder pct: 28.2%
- Time since launch: 32h
- MORI
- Price: 0.00000211 ETH
- 5m change: +4.2%
- 1h change: +12.3%
- Net flow 5m: +3.03 ETH
- Volume 5m: 5.26 ETH
- Volume 1h: 24.16 ETH
- Unique traders 5m: 20
- Top 20 holder pct: 31.1%
- Time since launch: 18h
- LUMA
- Price: 0.00000166 ETH
- 5m change: +2.2%
- 1h change: +6.5%
- Net flow 5m: +0.76 ETH
- Volume 5m: 3.69 ETH

- Volume 1h: 15.73 ETH
- Unique traders 5m: 11
- Top 20 holder pct: 66.8%
- Time since launch: 18h
- KIRO
- Price: 0.00000091 ETH
- 5m change: -4.7%
- 1h change: -9.3%
- Net flow 5m: +1.07 ETH
- Volume 5m: 4.34 ETH
- Volume 1h: 14.84 ETH
- Unique traders 5m: 14
- Top 20 holder pct: 32.9%
- Time since launch: 32h
- TAVO
- Price: 0.00000148 ETH
- 5m change: +7.3%
- 1h change: +4.5%
- Net flow 5m: +0.34 ETH
- Volume 5m: 1.50 ETH
- Volume 1h: 4.47 ETH
- Unique traders 5m: 4
- Top 20 holder pct: 58.2%
- Time since launch: 2h

ACTIVE STRATEGIES (CURRENT ONLY)

No active strategies.

ACTIVE SETTINGS

- Trading Activity (TA): 5 / 5 - How often you trade when there is fresh edge (1=very patient, 5=highly active). TA does NOT require a trade every tick.
- Asset Risk Preference (Risk): 2 / 5 - Which tokens you consider (1=prefer least volatile available, 5=embrace high volatility). Risk determines which tokens to consider, not whether to trade at all.
- Trade Size (Size): 3 / 5 - Maximum position sizing per trade (1=up to ~15%, 2=up to ~30%, 3=up to ~50%, 4=up to ~70%, 5=up to ~90% of available ETH). These are UPPER BOUNDS, not minimums.
- Holding Style (Hold): 2 / 5 - Minimum hold time before considering an exit (1=about ~30 minutes minimum; 2=about ~1 hour; 3=hours; 4=many hours; 5=days).
- Diversification (Div): 4 / 5 - Portfolio spread (1=concentrated in 1-2 tokens; 2=focused 2-3; 3=balanced 3-5; 4=spread 4-6; 5=wide 5+).

PORTFOLIO CONTEXT

- ETH: Balance: 0.240000
- No current token holdings.

CONSTRAINTS

- Max Trade Amount (Percent): 100.00% of available ETH

PRICE IMPACT LIMITS (max 1500 bps)

- NERA: BUY max 100.00% of ETH
- VEXA: BUY max 100.00% of ETH
- MORI: BUY max 100.00% of ETH
- LUMA: BUY max 100.00% of ETH
- KIRO: BUY max 100.00% of ETH
- TAVO: BUY max 100.00% of ETH

Respond with the single best action for this tick: buy, sell, or observe.